Unit 2 – Customer, Improvement and Quality Tools: Priority-Ordered Exam Notes

19MEE347 Total Quality Management · Built from your Unit 2 slides (139 slides) and your Unit 2 priority table

How to use these notes

The notes follow your priority list, but inside each section I order topics by **study value** (numericals first, because they are the most reliable marks). Importance tags:

- 🔴 **VERY IMPORTANT** – learn fully and practise writing/solving
- 🟡 **MODERATELY IMPORTANT** – key points plus one example
- 🟢 **JUST READ** – skim once

Mark weights are **my estimates** (Unit 2 is expected to carry roughly 48–55 marks of the 100, with about 13–20 of those numerical). They are not guaranteed exam questions.

⚠️ Check these before you rely on the slides

I re-did the slide calculations and found slips. Use the corrected values in the exam:

1. **X-bar/R worked problem (slide 54, same data as quiz Set A):** the slide says the average of the means is 44.6. The ten means add up to 440, so it is **44.0**. Correct limits are **40.694 to 47.306**, not 41.311 to 47.689. The conclusion is the same (points 35, 37, 49, 51 fall outside).
2. **p-chart example (slide 73):** the slide shows UCL = 0.066 and LCL = 0.036. That 0.066 is only the 3σ term. With p-bar = 0.0505 the correct limits are **UCL ≈ 0.116 and LCL = 0**. The process is in control.
3. **X-bar/R example 2 (slide 66):** the labels D3 and D4 are swapped in the R-chart lines. D4 is the upper-limit factor and D3 the lower-limit factor.
4. Where a question gives A2, D3, D4 (for example A2 = 0.58 for n = 5), use the given values, not the table values (0.577).

If your teacher marks with the slide numbers, show your formula and steps clearly so you can defend your answer.

Priority map

#	Topic	Importance	Type	Est. weight	Share of your Unit 2 time
A1	Control charts for variables (X-bar, R) PG 70	🔴 VERY IMPORTANT	Numerical	8 marks	17%
A2	Control charts for attributes (p, np, c, u)	🔴 VERY IMPORTANT	Numerical	5 marks	15%
A3	Process capability (Cp, Cpk)	🔴 VERY IMPORTANT	Numerical	3 marks	8%
A4	Seven tools of quality	🔴 VERY IMPORTANT	Concept	8 marks	12%
A5	Statistical fundamentals	🔴 VERY IMPORTANT	Concept + numerical foundation	supports A1–A3	8%
B1	Six Sigma and DPMO	🟡 MODERATE	Concept + possible numerical	6 marks	8%
B2	Benchmarking	🟡 MODERATE	Concept	6 marks	6%
B3	New seven management tools	🟡 MODERATE	Concept	6 marks	6%
B4	Customer satisfaction	🟡 MODERATE	Concept	6 marks	6%
B5	Customer retention	🟡 MODERATE	Concept	3 marks	3%
B6	Continuous process improvement / PDCA	🟡 MODERATE	Concept	4 marks	4%
C1–C4	Employee involvement, performance appraisal, supplier partnership, performance measures	🟢 JUST READ	Concept	short notes	7%

PG 48

PG 35

SECTION A – 🔴 VERY IMPORTANT

A1. Control charts for variables: X-bar and R charts 🔴 (highest numerical priority)

Why it matters: quiz Set A Q2 was exactly this; your slides spend about 10 slides on it.

When to use: the quality characteristic is **measured** (length, weight, etc.) and samples are small (the R chart suits n < 10). The X-bar chart watches the **process mean**; the R chart watches **variation within samples**. The limits rest on the normal distribution.

Formulas (from your slides):

```
X-bar_i = (sum of readings in sample i) / n          R_i = max − min of sample
i
X-double-bar = (sum of sample means) / g             R-bar = (sum of ranges) /
g      (g = number of samples)

X-bar chart:  CL = X-double-bar
              UCL = X-double-bar + A2 × R-bar
              LCL = X-double-bar − A2 × R-bar
R chart:      CL = R-bar
              UCL = D4 × R-bar
              LCL = D3 × R-bar
```

Factor table (from your slide):

n	2	3	4	5	6	7	8	9	10
A2	1.880	1.023	0.729	0.577	0.483	0.419	0.373	0.337	0.308
D3	0	0	0	0	0	0.076	0.136	0.184	0.223
D4	3.267	2.574	2.282	2.114	2.004	1.924	1.864	1.816	1.777
d2	1.128	1.693	2.059	2.326	2.534	2.704	2.847	2.970	3.078

Worked problem (quiz Set A, corrected): 10 samples of size 5; A2 = 0.58, D3 = 0, D4 = 2.11. Means: 43, 49, 35, 44, 45, 37, 51, 46, 43, 47. Ranges: 5, 6, 5, 7, 6, 4, 8, 6, 4, 6.

Step	Calculation	Result
X-double-bar	440 ÷ 10	44.0
R-bar	57 ÷ 10	5.7
X-bar UCL	44 + 0.58 × 5.7	47.306
X-bar LCL	44 − 0.58 × 5.7	40.694
R UCL	2.11 × 5.7	12.027
R LCL	0 × 5.7	0

Conclusion to write: on the X-bar chart, samples 49 and 51 lie above the UCL and 35 and 37 lie below the LCL, so the **mean is out of statistical control**. All ranges lie between 0 and 12.027, so the **variability is in control**. Assignable causes affecting the process average should be found and removed, then the limits recalculated.

Second slide example (n = 4, 20 samples): sum of means = 232, so X-double-bar = 11.6; R-bar = 4.15; A2 = 0.729 gives UCL = 14.63 and LCL = 8.57; D4 = 2.282 gives UCL(R) = 9.47 (about 9.5); D3 = 0 gives LCL(R) = 0.

Tests for out of control (from your slides):

1. One or more points outside the control limits
2. 9 points in a row above or below the centre line
3. 6 points in a row steadily increasing or decreasing
4. 14 points in a row alternating up and down

Solving template (use every time): (1) list data and given constants; (2) compute the two centre lines; (3) compute both pairs of limits; (4) compare every point; (5) write the conclusion in words.

A2. Control charts for attributes: p, np, c and u 🔴 (numerical)

Why it matters: quiz Set B Q2 was a p-chart; your slides have worked examples for all four.

Attribute data is counted, not measured. Choose the chart with this logic (slide flowchart):

Counting	Sample size	Chart	Plots	Distribution
Defectives (item is good/bad)	Variable or constant	**p**	Fraction defective	Binomial
Defectives	Constant	**np**	Number of defectives	Binomial
Defects (many per item)	Constant	**c**	Number of defects	Poisson
Defects	Variable	**u**	Defects per unit	Poisson

Formulas (from your slides):

```
p chart:   p-bar = (total defectives) / (total items inspected)
           UCL = p-bar + 3√[p-bar(1 − p-bar)/n]     LCL = p-bar − 3√[p-bar(1
− p-bar)/n]
           (varying size: use n_i for each sample; slide says n should be at
least 50)
np chart:  CL = n·p-bar
           UCL = n·p-bar + 3√[n·p-bar(1 − p-bar)]   LCL = n·p-bar − 3√[n·p-
bar(1 − p-bar)]
c chart:   c-bar = (total defects) / (number of samples)
           UCL = c-bar + 3√c-bar                  LCL = c-bar − 3√c-bar
u chart:   u-bar = (total defects) / (total units inspected)
           UCL = u-bar + 3√(u-bar/n_i)             LCL = u-bar − 3√(u-
bar/n_i)
```

Rule: if an LCL comes out negative, take LCL = 0.

Worked results from your slides (re-checked):

Example	Key numbers	Result
p chart, 20 CD samples of 100	total defective = 101; p-bar = 101 ÷ 2000 = 0.0505; 3σ = 3√(0.0505×0.9495/100) = 0.0657	UCL ≈ 0.116, LCL = 0 (raw −0.015); all points ⩽ 0.08, so **in control** (corrected from the slide)
np chart, bulbs, n = 200	n·p-bar = 5.3	UCL = 5.3 + 3√(5.3×0.9738) = 12.03; LCL = −1.5, so 0
c chart, TV sets, 20 samples	total defects = 109; c-bar = 5.45	UCL = 5.45 + 3√5.45 = 12.45; LCL = −1.55, so 0
u chart, motorbikes	u-bar = 0.417	Limits vary with n_i; the slide chart flags sample 8 as out of control

Quiz Set B (corrected data 8, 6, 7, 10, 5, 9, 4, 8, 6, 7; n = 100): total = 70, p-bar = 0.07; σ = √(0.07×0.93/100) = 0.0255; UCL = 0.1466; LCL = −0.0066, so 0. All sample fractions (0.04 to 0.10) lie inside, so the process is **in statistical control**.

Common mistakes: forgetting to divide the defectives by n for the p values you plot; using (1 − p-bar) wrongly in np; not setting a negative LCL to zero; choosing c when sample size varies (that is a u chart).

A3. Process capability: Cp and Cpk 🔴 (numerical)

Why it matters: syllabus topic, lecture 24, and the natural follow-on to control charts. **Your slides do not include process capability**, so this section is from standard TQM textbooks; confirm the exact formulas with your class notes.

Idea: a process can be in control and still not meet specifications. Process capability compares the **spread of the process (6σ)** with the **specification width**. Always check the process is in statistical control first.

```
Estimate of sigma:    σ = R-bar / d2                (d2 from the factor
table)
Cp  = (USL − LSL) / (6σ)                          (potential capability;
ignores centring)
CPU = (USL − μ) / (3σ)     CPL = (μ − LSL) / (3σ)
Cpk = min(CPU, CPL)                               (actual capability;
includes centring)
```

Interpretation (common convention): Cp or Cpk below 1.00 means not capable; 1.00–1.33 marginal; 1.33 or higher capable. Cpk ⩽ Cp always; Cpk = Cp only when the mean is exactly at the centre of the specification.

Practice example (my own): USL = 10.5, LSL = 9.5, μ = 10.1, σ = 0.15. Cp = 1.0 ÷ 0.90 = 1.11. CPU = 0.4 ÷ 0.45 = 0.89; CPL = 0.6 ÷ 0.45 = 1.33; Cpk = **0.89**. Comment: the spread would just fit (Cp 1.11), but the process is off-centre toward the upper limit, so it is not capable until the mean is shifted.

A4. Seven tools of quality 🔴

Why it matters: syllabus topic (lecture 19), in your priority table, and a classic 8-mark theory question. Your slides name the seven below. (Many textbooks list a **flow chart** instead of stratification, so mention it if asked for alternatives.)

#	Tool	What it does	Key facts to write
1	Check sheet	Collects data in real time where it is generated	Quantifies problem types, defects by type and by location; data may be quantitative or qualitative
2	Fishbone (cause-and-effect, Ishikawa)	Finds possible root causes and sorts them into categories	Named after Kaoru Ishikawa; categories 6M or 5M + 1E; structures brainstorming

#	Tool	What it does	Key facts to write
3	Histogram	Bar graph of how often each value occurs	Introduced by Karl Pearson; shows pattern and extent of variation, conformance to specification, and whether the process is in control
4	Pareto chart	Bars in descending order plus a cumulative line	80-20 rule (Vilfredo Pareto): about 80% of problems come from 20% of causes, the “vital few”; focus on the biggest causes
5	Control chart	Time plot with centre line, UCL and LCL	Shewhart chart; shows whether a process is stable and capable; limits come from historical data
6	Scatter diagram	Plots two variables to show correlation	Strong or weak, positive or negative relationship between cause and effect
7	Stratification	Splits data into sub-groups (by worker, supplier, department, shift)	Reveals which group a problem comes from; diagnoses and localises problems

Memory hook: **C-F-H-P-C-S-S** (Check, Fishbone, Histogram, Pareto, Control, Scatter, Stratification).

Pareto mini-example (my own): defect counts A 45, B 25, C 15, D 10, E 5 give cumulative 45%, 70%, 85%, 95%, 100%. A and B (two of five causes) produce 70% of defects, so fix those first.

A5. Statistical fundamentals 🔴 (concept + numerical foundation)

Your slides have no separate section on this, so these notes are from standard texts. They support everything in A1–A3, so learn them fast.

- **Data types:** variable (measured) vs attribute (counted).
- **Central tendency:** mean = $\Sigma x \div n$; median = middle value; mode = most frequent value.
- **Dispersion:** range R = max − min; variance $s^2 = \Sigma(x - \text{mean})^2 \div (n - 1)$; standard deviation s = √variance.

- **Normal distribution:** symmetric bell curve with mean = median = mode. Area within ±1σ = 68.26%, ±2σ = 95.46%, ±3σ = 99.73%. Standardise with z = (x − μ) ÷ σ.
- **Why control limits are 3σ:** about 99.73% of normal data falls inside, so a point outside is unlikely to be chance. Sample means have a smaller spread, σ ÷ √n, which is the basis of the X-bar limits.
- **Distributions behind the charts (your slides):** binomial for p and np, Poisson for c and u, normal for X-bar.
- **Variation:** common (chance) causes are built into the system; special (assignable) causes are what control charts detect (see Deming, Unit 1).

Mini numerical (my own): data 10, 12, 11, 13, 14 give mean 12, range 4, $s^2 = (4 + 0 + 1 + 1 + 4) \div 4 = 2.5$, s = 1.58. If μ = 50 and σ = 2, then P(X > 54) has z = 2, about 2.28%.

SECTION B – 🟡 MODERATELY IMPORTANT

B1. Six Sigma and DPMO 🟡

From your slides:

- Six Sigma is a disciplined, statistical, data-driven approach for eliminating defects in a product, process or service; it controls variation and replaces guesswork with statistics.
- Six Sigma quality means long-term defects below **3.4 per million opportunities (DPMO)**.
- Origin: Motorola in the 1980s (timeline slide mentions the name coined in 1986, Harry's ±1.5σ shift in 1987, and the 1988 Baldrige award).
- Aims to reduce **time, defects and variability**. Objectives: improve customer satisfaction, reduce cycle time, reduce errors, improve quality, eliminate defects.
- **DMAIC** (improving an existing process): Define, Measure, Analyse, Improve, Control. **DMADV** (creating a new product or process, also called DFSS): Define, Measure, Analyse, Design, Verify.
- Sample tools: Define – project charter, SIPOC, voice of customer; Measure – process map, control charts, process capability, Pareto; Analyse – fishbone, 5 Whys, FMEA, regression; Improve – DOE, brainstorming, 5S, TPM; Control – control charts, mistake-proofing, standard work.
- Belts: White, Yellow, Green, Black, Master Black, plus the Champion (executive leader who drives the initiative).

- Benefits: higher productivity, better quality, less waste, lower operating cost and variation, higher customer satisfaction, better employee engagement.

DPMO numerical (standard formula, not in your slides):

```
DPU  = defects / units
DPMO = defects / (units × opportunities per unit) × 1,000,000
Yield = 1 − DPMO / 1,000,000
```

Typical sigma levels (with the 1.5σ shift): 3σ = 66,807 DPMO; 4σ = 6,210; 5σ = 233; 6σ = 3.4. Example (mine): 50 units, 8 opportunities each, 20 defects: DPMO = 20 ÷ 400 × 10^6 = 50,000, a sigma level of about 3.1.

B2. Benchmarking 🟡

- **Definition:** comparing the cost, cycle time, productivity or quality of a process with an industry standard or best practice; a systematic way to measure yourself against the best, learn how they do it, and close the gap. It is a tool for continuous improvement.
- **5 steps:** choose what to benchmark → choose best-in-class partners → gather their data → compare to find gaps → adopt their processes.
- **6-step process:** decide what to benchmark; understand current performance; plan; study others; learn from the data; use the findings.
- **Importance:** improves processes, measures past performance, shows how competitors work, raises efficiency and lowers cost, improves quality and customer satisfaction.
- **Types (6):** process, financial, performance, product, strategic, functional.
- **Industry models in your slides:** AT&T 12 steps (identify clients → … → analyse data → integrate actions → take action → continue improvement) and Xerox 10 steps (identify what to benchmark; identify comparative organisations; collect data; determine the performance gap; project future performance; communicate findings; set functional goals; develop action plans; implement and monitor; recalibrate benchmarks). Learn the Xerox 10 steps; they are the easier list.

B3. New seven management tools 🟡

#	Tool	One-line purpose
1	Affinity diagram	Groups large amounts of unorganised ideas into natural categories (steps: capture, group, label)

#	Tool	One-line purpose
2	Activity network (arrow) diagram	Shows task order, best schedule, critical path and effect of delays
3	Process decision program chart (PDPC)	Identifies what might go wrong in a plan so countermeasures can be prepared
4	Prioritisation matrices	L-shaped matrix comparing options against criteria to choose the best
5	Matrix data analysis	Mathematical analysis of a matrix, often replaced by a prioritisation matrix
6	Tree diagram	Breaks broad goals into finer levels of detail, from general to specific
7	Interrelationship diagram	Arrows show cause-and-effect links among aspects of a complex issue

Your slides also show the **matrix diagram** (relationship between two or more groups, with strength); some textbooks list it in place of matrix data analysis. Learn both names.

When to use: affinity after brainstorming; interrelationship after affinity or fishbone; tree to turn issues into actions; arrow diagram when the schedule is critical.

B4. Customer satisfaction 🟡

- **Idea:** customer satisfaction is a subjective, dynamic part of the qualitative pillar of TQM; organisations use it as the measure of quality.
- **Customers:** internal (departments inside the company) and external (end users, purchasers, suppliers). Internal customer questions: what do you need from me; what do you do with my output; are there gaps?
- **Teboul model:** total satisfaction occurs when the company's offer matches the customer's need (the two circles overlap fully).
- **Kano model: Must-be** (basic, expected – e.g. clean seats on a bus), **More is better** (performance – free Wi-Fi), **Exciters/delighters** (unexpected – free streaming). Use it to prioritise what to improve and how to differentiate.
- **Good vs excellent service:** good service meets expectations; excellent service gives a little more; bad service falls short.
- **Customer perception of quality (ASQ survey):** performance, features, service, warranty, price, reputation.

- **Identifying needs:** surveys and market research, tracking complaints, customer meetings, competitor information, visits, employees with customer knowledge.
- **Complaints:** an expression of dissatisfaction; used to discover dissatisfaction and needs, learn priorities, compare with competitors and find improvements.
- **Feedback tools:** comment cards, questionnaires, post-transaction surveys, feedback cards, social media, toll-free numbers, customer visits. Feedback must be sought continuously, not once.
- **Why customers leave (your slide):** poor service 69%, product dissatisfaction 13%, better prices elsewhere 9%, others 9%.
- **Service organisations:** set key performance indicators and SMART targets, form teams, review weekly, standardise and sustain.

B5. Customer retention 🟡

Not covered in your slides (CDP lecture 13 pairs it with Kano), so this is standard-text content; check against your class notes.

- Retention means keeping existing customers through satisfaction plus loyalty. Retaining is cheaper than acquiring, and satisfied customers repeat-buy and recommend.
- **Retention rate = (customers at end − new customers) ÷ customers at start × 100.**
- Ways to improve: deliver consistent quality, resolve complaints fast, build relationships, seek feedback, and reward loyalty.
- Link to the slides: most customers leave because of poor service, so service quality drives retention.

B6. Continuous process improvement and PDCA 🟡

- **Modes:** detecting mode (inspection and firefighting) vs **preventive mode** (do it right the first time; use feedback to correct and improve). Variation is the major enemy.
- **PDSA cycle (Deming):** Plan → Do → Study → Act, repeated.
- **Model for improvement (3 questions):** what are we trying to accomplish; how will we know a change is an improvement; what change can we make that will result in improvement?
- **6-step problem solving:** identify the problem as an opportunity; analyse root causes; develop cost-effective solutions; implement system-wide; study the results; standardise.
- **Improvement strategy:** control the process (SPC) → optimise it (DOE) → change the system (hardware, software, brainware) → new system.

- **Why collect data:** objective information, effect of change, whether gains are retained, staff involvement, economic impact.
- **Culture:** positive and constructive, drive out fear, continuous feedback system, respond to customers promptly, standardise and keep improving.

SECTION C – 🟢 JUST READ

Your slides do not cover these four topics, so the points below are standard textbook content. Skim once; they are most likely short notes or optional questions.

- **C1 Employee involvement:** empowerment, teamwork, motivation (Maslow, Herzberg), suggestion schemes, recognition and reward. Benefit: people closest to the work know the problems; links to Deming points 8, 9 and 12.
- **C2 Performance appraisal:** purposes are feedback, development, pay and promotion; ideally uses team and customer input. Deming criticised annual merit rating as one of the seven deadly diseases (Unit 1).
- **C3 Supplier partnership:** long-term relationships with fewer suppliers, selection and rating on quality (not price alone), shared goals. Links to Deming point 4.
- **C4 Performance measures:** measure what matters to customers and the process, such as defect rate, cycle time, on-time delivery, customer satisfaction and cost of quality. They are set by the quality council.

Likely question skeletons (estimates, not guaranteed)

Likely question	Marks	Fast answer plan
X-bar and R limits from data + comment	8	A1 template; state both charts' conclusions
p / np / c / u chart limits + comment	5–6	A2 formulas; LCL negative → 0
Cp, Cpk for given specs, or sigma from R-bar/d2	3–5	A3 formulas + interpretation
Seven tools of quality	8	A4 table, one line each
Six Sigma concept, DMAIC, DPMO	6	B1
Benchmarking: process, benefits, types	6	B2
New seven management tools	6	B3 table

Likely question	Marks	Fast answer plan
Kano model and customer satisfaction	6	B4
Short notes (retention, CPI/PDCA, employee involvement, etc.)	3–4 each	B5, B6, Section C

45-minute last-minute plan

1. 15 min – Redo quiz Set A (corrected) and one other X-bar/R problem; memorise the formulas and factor table
2. 10 min – Redo quiz Set B (corrected), then one c-chart and one np-chart
3. 5 min – Cp and Cpk formulas and the practice example
4. 8 min – Seven tools table and Pareto example
5. 5 min – Six Sigma (DMAIC, 3.4 DPMO) and the benchmarking Xerox steps
6. 2 min – Skim B3 and Section C titles

Notes on source and limits

- Content comes from the text and key diagrams of your Unit 2 PDF. Items marked “my own example”, “standard” or “not in your slides” are my additions. Check them against your class notes.
- I viewed the control-chart, Six Sigma and benchmarking slides as images. I did not inspect every illustration (for example the Kano and Teboul diagrams or the windshield story in the Six Sigma slides).
- Importance levels follow your priority table; the mark estimates are mine.